

3D Printing Procedures

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1. Introduction

This document outlines the procedures for making use of 3D printing for skripsie and postgraduate use. Firstly, some important decisions to be made when considering 3D printing are identified. Thereafter, the specifications of the available printer and printing process are discussed and the procedure for using the printing facilities is described. This document is an adaptation of the guidelines proposed by Swart (2016).

2. Suitability for 3D printing

There are many aspects to consider when designing a component, and several of these aspects directly influence the manufacturing method that will be required. Some points to consider before deciding on 3D printing are:

- Can the component be purchased off the shelf?
- Can the component easily be manufactured through another available process?
- How many parts will be manufactured?
- To what loading, use and environmental conditions (e.g. forces, “wear and tear”, temperature, etc.) will the part be subjected?
- 3D printing is not always a quick way out of a tricky design or a remedy for time constraint issues.
- Do not design a detailed CAD model and then decide on a manufacturing process (i.e. design for manufacture, not the other way around!).
- Can the cost of manufacture be justified by the intended function of the component? An educated guess regarding costing is possible by using the component volume and other specifications given in this document.
- If weight constraints are of importance, do not default to 3D printing regardless of the shape of the component.

3. Printing specifications

3.1. Printing process

A majority of the available printers make use of Fusion Deposition Modelling (FDM), also known as Plastic Jet Printing (PJP). FDM is the process of melting plastic filament fed from a spool and depositing the molten plastic to the build surface through a small nozzle. As the plastic cools, it hardens and fuses to the underlying layer of material.

In comparison to other printing processes, FDM printers have much lower maintenance, running and material costs. The materials used by FDM printers are also more durable and tolerant of ambient temperatures than other common processes. Furthermore, FDM printers have much quicker build times than alternative printing processes. The primary drawback to FDM printing is that the accuracy is considerably lower than some other common printing processes, which makes it less suitable for complex parts.

The available printers primarily use Polylactic Acid (PLA), a thermoplastic polyester, as printing material. Some of the most important material properties of PLA are presented in Table 1.

Table 1: Material properties for PLA (from MakeItFrom.com (2021)).

Tensile strength	50 MPa
Melting temperature	160 °C
Density	1.3 g/cm ³
Elastic modulus	3.5 GPa
Elongation	6 %

The facility has one resin printer which makes use of a UV light source that cures fine layers of resin, slowly building up the part. Similarly to FDM, the resin printer creates the part from the bottom up, however the part is suspended from the build plate in the resin printer. The accuracy of this printer is higher than that of the FDM printers, however the build volume is limited, the consumables are more expensive and the process does take longer due to the post processing procedures of cleaning and curing.

3.2. Printers

The facility has the following FDM printers: four Bambu Lab P1S (Figure 1) and two Creality CR10S Pro V2 (Figure 2). The facility also has one resin printer, a Creality Halot Sky (Figure 3). The printer specifications are summarised in Table 2, Table 3 and Table 4.



Figure 1: Bambu Lab P1S printer (Bambu Lab (2023))



Figure 2: Creality CR10S Pro V2 (techgearlab.com)



Figure 3: Creality Halot Sky (<https://www.creality.com/goods-detail/creality-halot-sky-resin-3d-printer>)

Table 2: Bambulab P1S specifications (from Bambu Lab (2023)).

Nett build volume	256 mm x 256 mm x 256 mm ($l \times w \times h$) (250x250x250 usable)
Filament diameter	1.75 mm
Layer thickness	0.08-0.3 mm
Accuracy	< 0.2 mm
Print speed	200-300 mm/s
Build time	Highly dependent on component size, layer height and <i>infill</i> density ¹ .

Table 3: Creality CR10S Pro V2 specifications (from Creality (2020)).

Nett build volume	300 mm x 300 mm x 400 mm ($l \times w \times h$) (295x295x390 usable)
Filament diameter	1.75 mm
Layer thickness	0.02-0.3 mm
Accuracy	< 0.2 mm
Print speed	30-60 mm/s
Build time	Highly dependent on component size and <i>infill</i> density ¹ .

¹ *Infill* density refers to the degree to which the component's inner volume is reduced by creating honeycomb (or similar) support structures instead of producing a solid part.

Table 4: Creality Halot Sky specifications (from Creality (2022)).

Nett build volume	192 mm x 120 mm x 200 mm (<i>l x w x h</i>)
Resin type	UV curable 405 nm (Creality standard resin)
Layer thickness	0.01-0.2 mm
Accuracy	< 0.1 mm
Print speed	1-4 s/layer
Build time	Mostly dependant on part height and layer curing time.

3.3. Design for printing

Special care should be taken in the design of parts to make them suitable for 3D printing and to promote high quality builds. The most important design aspects are discussed in this section. Note that most of the considerations in this document are specified around FDM printing, however, they are still largely applicable to resin printing. Supports are usually required in resin printing as they are in FDM printing. Furthermore, the part is frequently orientated at a 45 ° angle. This is done to reduce the sticking force between the cured resin on the build plate and the FEP film in the resin vat. The part now being at an angle usually results in additional support that would not have been required when compared to FDM (as the parts are usually laid flat in FDM printing).

3.3.1. Support material

Slicer software, that convert the mesh files of a CAD design into machine code that is used to perform the print, provide little to no control over the shape and position of support structures. It comes down to including or excluding the support structure altogether (for FDM). The orientation of a component will often determine the need for support material, which is undesirable for FDM prints since support material degrades the surface finish of the component. Always make sure that the part is orientated in such a way that the need for support material is minimised, e.g. eliminating overhangs. The elimination of support material by changing a part's printing orientation is illustrated in Figure 4.

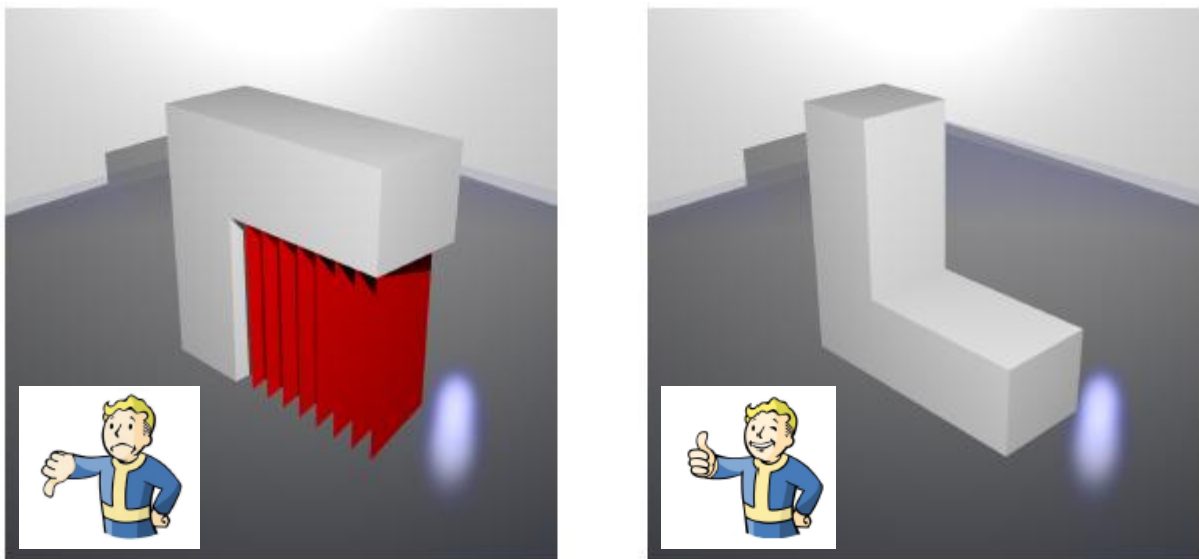


Figure 4: The effect of part orientation on the need for support material.

3.3.2. Accessibility

When designing a component that will require the generation of support structures, the designer needs to plan for the removal of the support. For example, a hollow cube with only one small hole in one of the sides will be nearly impossible to clean without extensive chemical post processing. Even when making use of chemicals for post processing some features cannot be cleared of the support material completely, such as in the case of a narrow bent tube.

3.3.3. Component splitting

Sometimes there exists no suitable orientation for the manufacture of a component. It is then more prudent to split a component into multiple parts and assemble them afterwards than to print the component as is, as is shown in Figure 5.

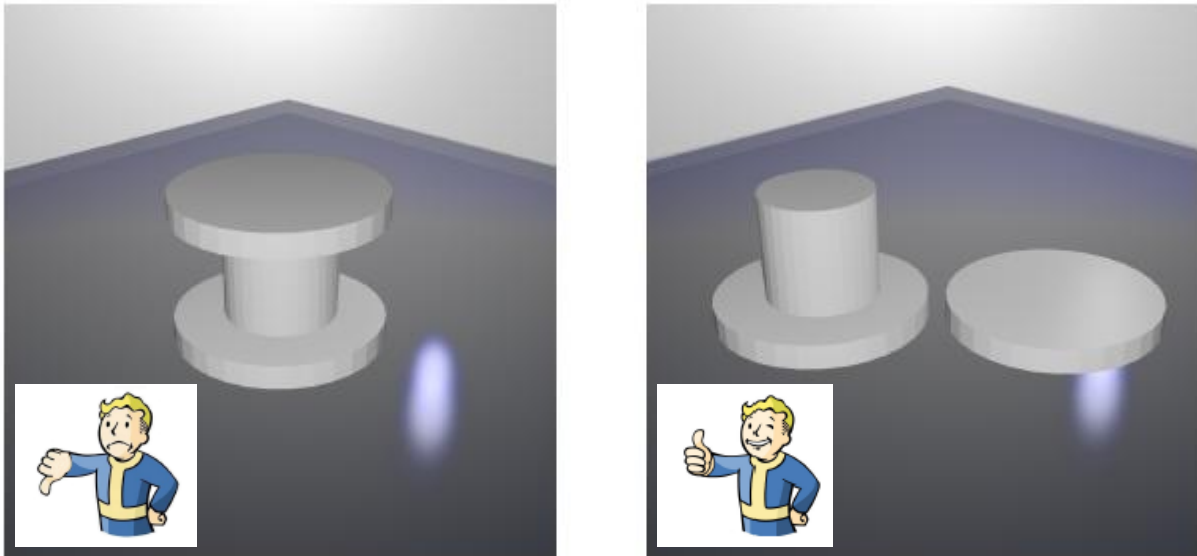


Figure 5: An example showing the need for component splitting.

3.3.4. Build and stress directions

When a load-bearing component is being designed, be sure that the direction of the load will be normal to the build plan. 3D printed parts can handle compression and tension loading (i.e. normal to the layers) much better than shear or torsional loading. The reason for this is that the component is weakest between the layers since the attachment between layers do not allow for full material strength.

3.3.5. Cooling rate

The rate of cooling for the deposited plastic can influence the quality of the print greatly. Temperature gradients, that exists when cooling does not occur uniformly throughout a part, induce thermal stresses that may cause warping. This is frequently encountered in large and high-density infill components. Some ways to minimise the risk of this happening are to avoid unnecessarily large volumes and avoiding stress concentration features (e.g. sharp corners).

In cases where the cooling rate is too low, smearing tends to occur as the extruder head jumps between printing areas. This situation is most common when the print area is very small, such as in the case of spire-like structures. Methods of reducing the likelihood of this occurring include creating extra thin wall structures that can be broken off the part after the print and providing a continuous printing path to ensure that the extruder will not jump between print areas, but rather transition between them smoothly. An example of this design consideration is shown in Figure 6.

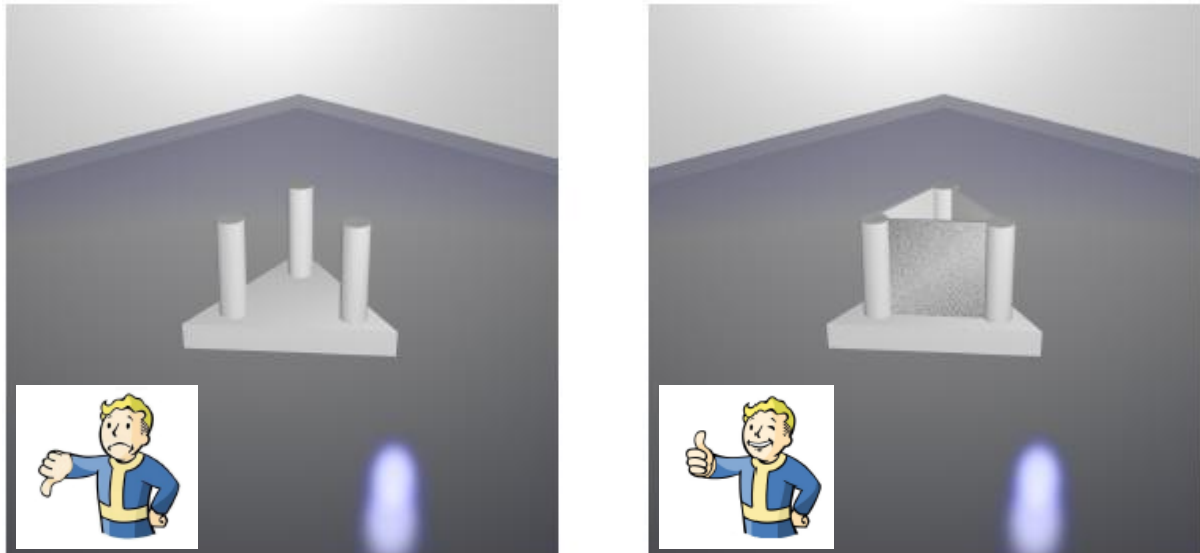


Figure 6: The insertion of support material between spire-like structures.

3.3.6. Wall thickness and clearance

A minimum wall thickness of 1 mm is suggested for FDM prints. Features with wall thicknesses this thin, should only be used for small parts or as supporting structures in larger prints. When designing components that will be assembled into other parts, tolerances should be built into the dimensions of the component.

When the component to be printed has holes, it is recommended that the holes diameters should be made at least 1mm less than the desired dimension and then drilled to the correct size after manufacture.

3.4. Post processing

Post processing for FDM prints is relatively simple. It mainly consists of support removal and manually cleaning or filing of any discrepancies in the part. The process for resin prints is similar, however the part is first washed in isopropyl alcohol, the support material is removed, and lastly, the part is cured in a curing station.

4. Procedure

The procedure that must be followed through the design, preparation and request stages, is outlined in this section.

4.1. Design and preparation

Parts can be designed in Inventor (or your preferred CAD package), as normal .IPT files. When the design is completed, the part must be saved or exported as a .STL file (.STEP is only

compatible with Bambu Studio). Ensure that the STL file is scaled correctly and it exported in high quality. Mesh quality issues are usually apparent on holes or curves. To get an idea of printing time and cost, the .STL file can then be imported to the Bambu Studio or Ultimaker Cura software (for FDM). The software can be downloaded from [Bambu Studio](#) and [Ultimaker Cura](#). Creality Halot Box software can be downloaded from [Creality Halot Box](#).

The printing software allows the designer to specify part orientation and printing settings. The generic settings typically used are tabulated in Table 5.

Table 5: Recommend settings for estimations in Cura and Bambu Studio software.

Software	Ultimaker Cura	Bambu Studio
Which Printer?	CR10S Pro V2 and Ender 3	P1S
Profile	Standard quality	0.2mm Standard @BBL X1C
Print setup	Default	P1S 0.4 Nozzle,
Layer Height	0.2 mm	0.2mm
Infill	20 % ²	20% ²
Generate support	✓	✓
Build plate adhesion	None	None

Based on the nature of the part and the printing settings, the slicing software provides three estimations – printing time, filament weight and filament length. Halot Box has similar estimates of resin volume and print time. The time restriction and rough estimation of printing cost, as shown in Table 6, must be considered.

² 20 % infill results in relatively strong parts – if additional strength is required, it should be motivated in the request. In this case, evaluate if 3D printing is a suitable process for manufacturing the part.

Table 6: Printing time and costs.

Maximum allowable printing time	No restriction, but long prints usually have a higher risk of failure. This is accounted for in the quotation system.
Printing cost	FDM, R50 setup fee per build plate. Approximately R15/hour, R370/kg for FDM. Resin, R200 setup fee per build plate. Approximately R18/hour, R1/ml of resin for resin prints.

4.2. Job requests

All 3D printing will be supervised by the facilitator, Elijah Stockhall (23648341@sun.ac.za). Ensure that your supervisor has reviewed your parts and drawings before creating a request. Printing jobs can then be requested by email, with the following structure:

To: 23648341@sun.ac.za Subject: 3D printing request ## .STL and .PDF drawing files attached##
Material type ³ : ## Rough outer dimensions of each part: ## QTY of each part: ## Specific orientations for the part to be printed in ⁴ : ## Specific print settings ⁵ : ## Supervisor name: ## Supervisor approval: Y/N Comments: ##

Note that for each part, engineering drawings that adhere to the requirements laid out in “Procedure for Workshop Work in MMW” (found on STEMLearn) must be supplied. The drawings must be supplied as PDF format. Please provide STL files (double check scaling and quality) for every part. If the job request entails multiple parts, attach all part files to one email. If necessary please specify per part settings otherwise one setting will be used for all.

³ We stock and can print PLA, PETG, ABS (ABS is likely to have warping issues on very large or tall parts) and resin. PLA is by far the easiest to print. It is possible to print other filaments, however this must first be discussed with Elijah Stockhall.

⁴ If no orientation is specified, the orientation which is easiest to print will be selected. If a text explanation of orientation is confusing please provide drawings. Note that most resin prints are printed at approximately a 45 ° angle to reduce the ‘sticking’ forces between the layer and the FEP film.

⁵ If no settings are specified, generic settings will be used, mainly, infill %, layer height and wall thickness. For resin prints, the standard Halot Box settings will be used in most cases. That is a 0.05 mm layer height and 100 % infill part.

4.3. Part returns and feedback

If the print request satisfies all the requirements, a quotation will be emailed to you. Note that for jobs under R300 your supervisor does not need to approve the quotation and printing will be started as soon as possible after the quotation was sent. For jobs costing more than R300 or you are a postgraduate student, please have your supervisor approve the quotation and forward the approval email or approved quotation with the cost point supplied by your supervisor to 23648341@sun.ac.za. Once approved the printing is usually completed within a week and can be collected from Elijah Stockhall. If parts require major revisions, this will be communicated and the job request procedure must be restarted.

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